



Exploring EFL students' effective use of GenAI in writing: Interplay of self-regulation, critical thinking, growth mindset, general writing competence, and GenAI literacy

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Abstract

Understanding individual factors influencing EFL students' effective use of GenAI in writing is essential for advancing EFL writing instruction in the GenAI era. However, empirical research in this area remains limited. Grounded in Social Cognitive Theory (SCT), this study introduces a model that conceptualizes EFL students as agentic learners who dynamically coordinate personal resources—cognitive (self-regulation, critical thinking), motivational (growth mindset), technological (GenAI literacy), and linguistic (general writing competence)—to regulate their behavior and engage effectively in GenAI-assisted writing. Data were collected from 342 first-year EFL students who completed a semester-long college English course integrating GenAI tools into writing, using questionnaires and two writing tasks. The model was tested using partial least squares structural equation modeling (PLS-SEM). The findings indicated that 1) self-regulation, critical thinking, GenAI literacy, and general writing competence all significantly influenced effective GenAI use. Among these, self-regulation and critical thinking were the strongest predictors, followed by GenAI literacy, while general writing competence had a lower but still significant impact. 2) Growth mindset did not directly impact GenAI use but had an indirect effect through critical thinking and self-regulation. 3) General writing competence significantly predicted growth mindset and critical thinking, while critical thinking was a key determinant of self-regulation and GenAI literacy. These results suggest that cognitive factors are primary drivers of effective GenAI use, with GenAI literacy serving a supportive role. Unlike previous perspectives that highlighted writing competence and motivation as central, these factors now play a less significant role in the GenAI-assisted writing context. Theoretical and practical implications for writing pedagogy are discussed, including fostering cognitive skills, integrating GenAI literacy, re-conceptualizing traditional writing competence, sustaining adaptive motivation in the GenAI era, and addressing ethical concerns.

Keywords GenAI-assisted writing · Effective GenAI use · EFL students · Self-regulation · Critical thinking · Growth mindset · General writing competence · GenAI literacy

1 Introduction

The rapid advancement of Generative AI (GenAI), with tools like ChatGPT, has significantly enhanced AI-driven writing support. These large language models leverage reinforcement learning from human feedback to generate more accurate, objective, and contextually appropriate responses. This progress introduces new possibilities in writing support, including real-time feedback, improved linguistic accuracy, style refinement, and idea generation. As GenAI evolves, it is expected to revolutionize writing by making it more efficient, precise, and accessible (Feng Teng, 2024). This potential holds notable implications for writing education (Gilburt, 2023; Marzuki et al., 2023), especially in EFL contexts, where students must balance linguistic accuracy with self-expression in a second language. Consequently, GenAI-assisted EFL writing instruction has become a key focus in writing pedagogy research.

Existing research falls into four main strands. The first focuses on GenAI applications, particularly in automated assessment and tutoring, highlighting its effectiveness in essay scoring (Bucol & Sangkawong, 2024; Mizumoto & Eguchi, 2023) and comprehensive feedback (Barrot, 2023; Guo & Wang, 2024). The second examines the impact of GenAI on learners and teachers, reporting improvements in writing skills (Gozali et al., 2024), language development (Punar Özçelik & Yangin Eksi, 2024), motivation and engagement (Feng Teng, 2024), and teaching efficacy (Ghafour et al., 2024). The third explores EFL students' perceptions, revealing generally positive views on GenAI's ability to enhance clarity, coherence, and collaboration (Allen & Mizumoto, 2024; Boudouaia et al., 2024), although some concerns remain regarding its handling of context-specific needs (Punar Özçelik & Yangin Eksi, 2024). The fourth addresses concerns related to GenAI use, including plagiarism, academic integrity (Shiri, 2023; Zhai, 2023), over-reliance (Shen & Teng, 2024), and the risk of homogenization (Gozali et al., 2024; Yu, 2023).

Despite these potential benefits, concerns remain about whether EFL students can integrate GenAI effectively into their writing. To date, little research has explored how students might maximize these benefits by considering personal factors that influence effective GenAI use. As noted by Cadwell et al. (2016) and Yang and Wang (2019), user-centered research on human factors is crucial to understand technology's impact on learners. Given that EFL students are the primary users of GenAI-assisted writing, it is critical to examine how individual attributes predict effective engagement with GenAI.

To address this, the present study adopts Social Cognitive Theory (SCT) (Bandura, 1986, 2001, 2020) as its guiding framework. SCT highlights the reciprocal interplay among personal attributes (such as cognition, motivation, and beliefs), behavioral regulation, and environmental affordances in shaping agentic learning behavior. This theoretical lens is especially well-suited to GenAI-assisted writing, where learners must regulate their writing processes, critically evaluate and refine

GenAI-generated content, draw on their existing linguistic competence, and adapt to evolving technological features. Moreover, this framework resonates with findings in writing education research, which consistently emphasize the importance of cognitive, motivational, linguistic, and technological dimensions in learners' engagement with digital tools (Guo & Lee, 2023; Liu et al., 2024; Sullivan et al., 2024).

Drawing on these theoretical insights, this study identifies five key constructs—general writing competence, self-regulation, critical thinking, growth mindset, and GenAI literacy—that together form an SCT-based model of learner engagement in GenAI-assisted writing. General writing competence, as a core personal factor, reflects learners' domain-specific knowledge and skills. It provides the linguistic foundation needed to plan, evaluate, and integrate GenAI-generated content (Taskiran et al., 2024; Yasuda, 2011), and serves as a basis for purposeful action in writing tasks (Bandura, 2020). Self-regulation and critical thinking, aligned with SCT's behavioral regulation dimension, function as essential self-directed processes. The former supports planning, monitoring, and iterative revision (Kaplan et al., 2019; Liu et al., 2024), while the latter enables evaluative judgments about the relevance and accuracy of GenAI output—a process that Bandura (2001) describes as self-correction and functional adaptation (Guo & Lee, 2023). Growth mindset, as a personal motivational factor, shapes learners' beliefs about effort, persistence, and improvement. Students who view writing as improvable are more likely to engage meaningfully with GenAI, activating both self-regulation and critical thinking in line with the motivational influence of forethought (Tu, 2024; Xu & Wang, 2022). GenAI literacy refers not only to basic operational skills but also to a deeper understanding of how generative AI functions and the critical ability to apply it effectively in writing. This is evident in students' ability to design prompts, interpret AI feedback, and refine outputs to improve quality (Kim et al., 2025; Sullivan et al., 2024). Within the SCT framework, GenAI literacy reflects purposeful interaction with the environment, enabling learners to shape digital tools to serve learning goals—a form of proactive transformation rather than passive adaptation (Bandura, 2020). Importantly, it is also directly linked to behavioral regulation, providing the conceptual, procedural, and evaluative resources that underpin ongoing monitoring and self-reflection in technology-mediated writing. Together, these five constructs form an SCT-based model that positions EFL students as agentic learners who coordinate their personal, behavioral, and environmental resources—spanning cognitive, motivational, linguistic, and technological domains—to engage effectively in GenAI-assisted writing.

Building on the SCT-based framework proposed in this study (Fig. 1), this study explores how self-regulation, critical thinking, growth mindset, general writing competence, and GenAI literacy predict EFL students' effective use of GenAI in writing, as well as their interplay. Data were gathered from 342 EFL students enrolled in a semester-long college English course that integrated GenAI tools into writing instruction. The data collection involved questionnaires and two writing tasks—one with GenAI assistance and one without. Descriptive statistics and partial least squares structural equation modeling (PLS-SEM) were used to analyze the collected data. The findings shed light on how these factors interact to foster effective GenAI use, clarify underlying mechanisms, and offer pedagogical implications for integrating GenAI into EFL writing instruction.

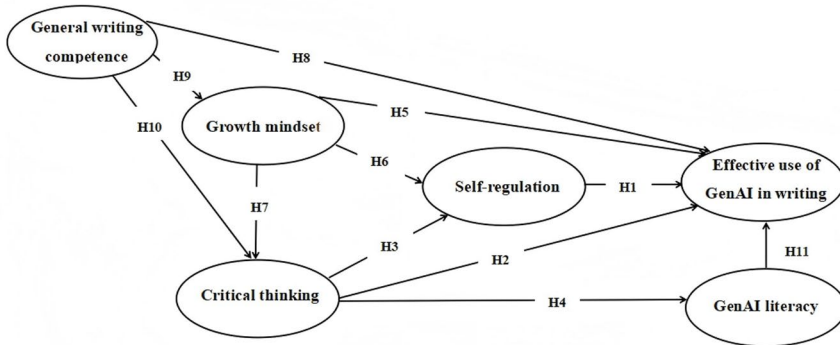


Fig. 1 Research model for EFL students' effective use of GenAI in writing

2 Relevant studies

2.1 GenAI-assisted EFL writing

Writing has transformed from a simple pen-and-paper activity into a dynamic, digitally driven process. Traditionally, it involved stages such as prewriting, planning, drafting, revising, and reflecting (Gezmiş, 2020), using physical tools like pens and typewriters that fostered deep, linear, and often solitary engagement with the text (Kawinkoonlasate, 2019). With technological advancement, writing evolved into a digitalized process, fundamentally reshaping how writers interact with text (Feng Teng, 2024). Early digital tools like word processors eased drafting and revision by eliminating physical constraints, while spell and grammar checkers enabled quicker error correction and greater attention to content. The increasing accessibility of online resources—such as dictionaries and research databases like Google Scholar and JSTOR—further shifted writing into a more informed and resource-driven activity (Putnam, 2001; Umamah & Cahyono, 2022). This evolution accelerated with the emergence of AI-powered writing tools. Applications like Grammarly and automatic writing evaluation systems now offer real-time feedback on grammar, tone, and clarity (Dewi, 2022; Ding & Zou, 2024), while tools like QuillBot assist in refining text and overcoming writer's block (Fitria, 2021). Machine translation tools and collaborative platforms such as Google Docs further support various writing stages and enable synchronous peer collaboration (Kitjaroonchai & Suppasetsee, 2021; Lee, 2020). These advances have made writing more interactive, collaborative, and resource-rich—a shift particularly significant for EFL learners who must master a second language while expressing complex ideas.

Amid the ongoing digital transformation, a more recent and transformative shift has emerged: the integration of GenAI into writing. Tools such as ChatGPT and Claude build on earlier technologies by offering not only automated assistance but also interactive, adaptive language support (Feng Teng, 2024; Wang & Dang, 2024). In EFL writing, GenAI enhances multiple stages of the process. During prewriting, it supports brainstorming, structured outlining, and idea development through dia-

logic prompting (Allen & Mizumoto, 2024; Boudouaia et al., 2024). In drafting, it enhances fluency by supplying example sentences, vocabulary, and genre-specific templates that help learners articulate complex ideas. During revision, it provides tailored feedback on grammar, structure, and tone, helping improve cohesion and coherence—key criteria for high-quality writing (Mizumoto & Eguchi, 2023). GenAI also promotes self-editing by guiding students through iterative revision cycles (Song & Song, 2023). In the reflection phase, it offers explanations for corrections and tools for paraphrasing and summarizing, encouraging students to critically explore and refine their language (Shen & Teng, 2024).

Despite the promising opportunities, GenAI-assisted EFL writing also raises concerns, particularly regarding over-reliance and academic integrity. One major issue is that students may become excessively dependent on GenAI-generated content, potentially hindering the development of their language competence and independent thinking. The ease of generating polished content also increases the risk of plagiarism and ethical misuse. Chomsky et al. (2023) referred to this as “high-tech plagiarism,” while Day (2023) warned that such dependence may undermine human creativity, intuition, and cultural sensitivity—essential elements in authentic writing.

In summary, while GenAI offers significant opportunities to enhance writing support in EFL contexts, its integration presents challenges. This dual nature underscores the need for further research to examine its pedagogical impact and to develop best practices for responsible and effective use in writing instruction.

2.2 GenAI-assisted EFL writing instruction

Given the potential benefits discussed previously, research on GenAI-assisted EFL writing instruction has gained increasing attention. Current literature generally follows four main strands.

The first line of research focuses on the application of GenAI in EFL writing classrooms, particularly for automated assessment and supplementary tutoring. For example, Barrot (2023) and Guo and Wang (2024) reported that GenAI tools serve as virtual tutors, offering writing suggestions and addressing grammar-related queries. Mizumoto and Eguchi (2023) demonstrated that GenAI-based feedback systems can provide automated diagnostic feedback on syntactic and rhetorical aspects, while Bucol and Sangkawong (2024) showed that GenAI-based automated essay scores closely match those of human raters. These findings underscore GenAI’s potential to enhance classroom instruction through personalized, timely feedback.

The second strand explores the impact of GenAI on students’ writing performance and engagement. Studies consistently indicate that GenAI can positively affect EFL learners’ writing quality. For instance, Gozali et al. (2024) and Punar Özçelik and Yangin Eksi (2024) found that GenAI-assisted revision improves grammar, lexical diversity, and coherence. Beyond linguistic gains, Feng Teng (2024) reported that instant AI-generated feedback boosts student engagement and reduces writing anxiety. Moreover, teachers benefit from GenAI by automating routine tasks such as proofreading, thereby improving teaching efficacy (Ghafouri et al., 2024). These findings highlight the need to investigate not only what GenAI can offer, but also how learner-internal factors influence the extent to which students benefit from such tools.

Another line of research has explored the perceptions of students and teachers regarding GenAI use. Overall, users report positive experiences, citing the usefulness of GenAI in clarifying ideas and improving text structure (Allen & Mizumoto, 2024; Boudouaia et al., 2024). However, some studies highlight limitations—particularly concerning cultural relevance and context-specific nuances (Punar Özçelik & Yangin Eksi, 2024). Although these studies shed light on user attitudes, few have linked these perceptions to specific learner attributes, such as critical thinking or self-regulation.

A final strand addresses the ethical challenges of using GenAI in EFL writing instruction. Research by Shiri (2023) and Zhai (2023) emphasizes the risk of misuse, including plagiarism, while Shen and Teng (2024) warn that excessive reliance on GenAI may undermine critical and independent thinking. Additionally, studies by Gozali et al. (2024) and Yu (2023) raise concerns about writing homogenization, noting that similar structures and vocabulary across users can stifle creativity and individual expression. These issues raise important questions about how to ensure effective and responsible use, especially through the cultivation of individual capacities.

Taken together, current research provides valuable insights into GenAI's classroom applications, impacts, user perceptions, and associated risks. However, the focus has largely been on tool integration and output quality, with limited attention to how learner-internal factors shape effective GenAI use. To address this gap, the current study adopts the SCT framework (Bandura, 1986, 2001, 2020) to systematically conceptualize the interrelations among the individual variables in the GenAI-assisted EFL writing context.

2.3 Individual factors in GenAI-assisted EFL writing: A SCT perspective

2.3.1 Social cognitive theory

SCT conceptualizes human functioning as a product of triadic reciprocal causation among three interacting dimensions: personal factors, behavioral regulation patterns, and environmental influences (Bandura, 1986, 2001, 2020). Personal factors include internal cognitive, affective, and motivational processes such as knowledge, beliefs, and goals. Behavioral factors refer to individual's observable actions and self-regulatory strategies that guide learning and goal attainment. Environmental influences involve external affordances and social contexts that offer feedback, opportunities, and constraints. These dimensions operate interactively, forming a dynamic system in which individuals both influence and are influenced by their actions and surroundings.

At the core of SCT is the concept of human agency—the capacity to intentionally regulate one's behavior and shape outcomes (Bandura, 1986). Agency is enacted through the integration of personal resources, behavioral strategies, and engagement with the environment. For example, knowledge and motivation enable learners to interpret tasks, set goals, and sustain effort (Bandura, 2001; Schunk & DiBenedetto, 2020). Behavioral regulation, such as planning, monitoring, and adjusting strategies, supports performance (Bandura, 1986, 2001). Meanwhile, purposeful interaction with environmental affordances (such as tools, feedback, and social input) allows

individuals to shape their learning context in ways that serve their intentions (Bandura, 2001; Nickerson, 2024). These agentic processes are embedded in SCT's triadic model and function together to support self-directed learning and adaptive development. Within this framework, learners are seen as proactive agents who coordinate cognitive, behavioral, and contextual resources to manage learning effectively.

This integrated framework offers a clear basis for understanding student engagement in GenAI-assisted EFL writing. In such contexts, students do more than receive GenAI-generated content; they must critically evaluate suggestions, revise iteratively, and incorporate external input into coherent, goal-oriented output. These tasks draw on multiple agentic capacities, including prior writing knowledge, sustained motivation, behavioral regulation, and real-time adaptation to technological affordances. Such demands align closely with SCT, which characterizes learners as intentional and self-regulating agents operating within dynamic environments. The triadic model is particularly well-suited to this context, as it accounts for the dynamic interaction among personal attributes, behavioral strategies, and environmental influences. Based on this, the next section examines five individual factors, each reflecting a distinct dimension of agentic functioning in GenAI-assisted EFL writing.

2.3.2 SCT-based individual factors in GenAI-assisted EFL writing

Guided by the SCT framework, this study examines five individual factors central to student engagement in GenAI-assisted EFL writing: general writing competence, self-regulation, critical thinking, growth mindset, and GenAI literacy. These factors correspond to key components of SCT's triadic model, including personal knowledge and skills, behavioral regulation, motivational beliefs, and interaction with environmental affordances. Collectively, they span four domains of learner engagement—linguistic, cognitive, motivational, and technological—and together form a coherent framework for understanding how EFL students interact with GenAI in writing tasks.

Linguistically, general writing competence represents a core personal factor in SCT, referring to the linguistic knowledge and task-specific skills that underpin agentic engagement in writing (Yasuda, 2011). It enables EFL students to produce coherent, well-structured texts and to evaluate and integrate GenAI-generated content. Prior research shows that students with stronger writing abilities perform better in revision, coherence maintenance, and the adaptive use of AI feedback (Gabrielatos, 2002; Nazarov & Kuvandikova, 2023). From the SCT perspective, such competence equips learners with the internal resources needed to use GenAI purposefully and strategically in writing tasks.

Cognitively, self-regulation and critical thinking correspond to SCT's behavioral-regulation dimension, emphasizing how learners plan, monitor, and adjust their performance in response to evolving task demands (Nickerson, 2024). Self-regulation facilitates goal-setting and ongoing control of writing behaviors, while critical thinking enables EFL students to assess and refine GenAI-generated content to ensure clarity and logical consistency. These capabilities exemplify the self-regulatory subfunctions of forethought, performance monitoring, and self-reflection—core mechanisms by which learners exercise agency in complex tasks (Baranowski et al., 2002). Recent studies using PLS-SEM provide empirical support: Jin et al. (2025) found that

higher self-regulation levels predicted deeper engagement with GenAI tools, while Zhou et al. (2024) reported that critical thinking positively influenced final writing quality. Together, these constructs serve as key cognitive mechanisms that help EFL learners manage their interaction with GenAI in strategic and adaptive ways.

Motivationally, growth mindset reflects EFL students' beliefs about effort and improvement, which are central to SCT's conception of personal agency. Rooted in Dweck's (2006) framework, it embodies the view that writing ability can be developed through practice. This belief fosters persistence, goal-setting, and openness to feedback, all of which support sustained engagement with GenAI tools (Chomsky et al., 2023; Yao et al., 2024). In SCT terms, growth mindset aligns with Bandura's notion of forethought, the motivational process through which learners anticipate outcomes and commit to purposeful action. While GenAI-specific research remains limited, a PLS-SEM study by Huang and Mizumoto (2024) found that students with stronger growth mindset showed greater confidence and initiative when using GenAI for writing. These findings support the role of growth mindset as a motivational driver of agentic engagement in GenAI-supported contexts.

Technologically, GenAI literacy is a multidimensional construct that captures students' capacity to engage purposefully with environmental affordances and coordinate behavioral regulation in GenAI-supported writing. It extends far beyond operational skills to include conceptual understanding of generative AI, strategic prompt engineering, critical and ethical evaluation of AI-generated content, and the ability to integrate useful suggestions into writing (Kim et al., 2025; Zhao et al., 2024). Students with high GenAI literacy are not passive recipients of technology but proactive agents who shape their environment to achieve learning goals—a process described in SCT as triadic reciprocal determinism (Bandura, 2020). Crucially, GenAI literacy functions as a bridge between environmental affordances and behavioral regulation: it enables students to actively identify and leverage the capabilities and limitations of GenAI tools, transforming these external resources into self-regulatory practices. By drawing on conceptual knowledge, procedural skills, evaluative judgment, and ethical awareness, students monitor the relevance, accuracy, and coherence of GenAI outputs, and reflect on and adjust their prompting strategies and writing decisions. Through this dynamic, agentic process, learners regulate their engagement with GenAI tools, exemplifying core SCT mechanisms such as ongoing monitoring and self-reflection in technology-mediated writing environments (Zhao et al., 2024). While research in EFL contexts is still emerging, existing empirical findings support this view. A PLS-SEM study by Kim et al. (2025) showed that higher GenAI literacy predicts deeper tool engagement and better writing outcomes, underscoring its critical role in supporting agency in digital writing environments.

Despite growing interest in GenAI-assisted EFL writing, research on individual factors in GenAI-assisted EFL writing remains limited. Most existing studies focus on broader technology-enhanced contexts and typically examine isolated variables. For example, Zhou et al. (2024) investigated self-regulation, while Kim et al. (2025) focused on GenAI literacy. Few have adopted a theoretically grounded approach to explore how multiple individual attributes interact to shape engagement. In particular, the dynamic relationships among linguistic, cognitive, motivational, and technological factors remain under-explored. Addressing these gaps, this study applies SCT

to examine how individual factors interact to shape EFL students' engagement with GenAI-assisted writing.

3 Research statement and questions

Research on how EFL students use GenAI effectively in writing is still in its early stages, with few studies examining the impact of individual factors and their interactions. Guided by SCT (Bandura, 1986, 2001, 2020), this study develops an SCT-based model that integrates five learner attributes: general writing competence, self-regulation, critical thinking, growth mindset, and GenAI literacy. These factors align with key components of SCT's triadic framework and collectively represent the linguistic, cognitive, motivational, and technological dimensions of agentic functioning in GenAI-assisted writing. Within this model, we propose several hypotheses to explore how these variables predict effective GenAI use. Accordingly, three research questions were formulated.

RQ1: Guided by SCT, which individual learner factors significantly predict EFL students' effective use of GenAI in writing?

RQ2: What are the structural relationships among the identified individual factors within the SCT-based model, and how do they predict EFL students' engagement with GenAI-assisted writing?

RQ3: What theoretical and practical implications do these relationships have for advancing our understanding of GenAI use in EFL writing contexts?

To model these complex interrelationships, Partial Least Squares Structural Equation Modeling (PLS-SEM) is employed—a method well-suited for exploratory research involving latent constructs and emerging technologies (Hair et al. 2013). While prior studies (e.g., Jin et al., 2025; Zhou et al., 2024) have applied PLS-SEM to GenAI-assisted writing, they typically focus on isolated variables. In contrast, this study adopts an integrative approach grounded in SCT to assess how multiple factors operate within a theoretically specified structure. Rather than exploring all possible interactions, this study focuses on their structural relationships as conceptualized within the SCT-based model. Evidence from other domains (e.g., Cai et al., 2025; Di et al., 2024) further supports the use of SEM-based methods for modeling complex behavioral processes, reinforcing the methodological rationale for our study.

4 Research model and hypotheses

Grounded in SCT (Bandura, 1986, 2001, 2020), this study develops a structural model to examine five individual factors influencing EFL students' effective use of GenAI in writing. The model includes 11 hypotheses that reflect theoretically and empirically supported relationships among these variables, drawing on prior research in technology-enhanced writing contexts. The following sections elaborate on each construct, along with its associated hypothesis and empirical justification.

4.1 Self-regulation

Self-regulation refers to an individual's cognitive ability to manage thoughts, emotions, and behaviors to achieve specific goals (Zimmerman, 2002). Within the SCT framework, it functions as a core mechanism of behavioral regulation that enables learners to exercise agency through forethought, performance monitoring, and self-reflection (Bandura, 2020; Onoda, 2012; Zimmerman & Kitsantas, 2014). In this study, self-regulation is defined as EFL students' ability to purposefully manage their use of GenAI tools throughout the writing process. This includes planning their interactions with GenAI tools, critically evaluating GenAI-generated content, and refining their writing strategies to enhance output quality. This deliberate behavioral control reflects SCT's emphasis on how learners actively regulate their engagement with external affordances in response to task demands and feedback.

Prior research has shown that self-regulated learners tend to achieve better writing outcomes through goal-setting, sustained attention, and adaptive strategy use (Kaplan et al., 2019). Empirical studies have also indicated that self-regulation enhances students' use of digital tools such as Grammarly and Google Docs (Ebadi & Rahimi, 2019; Li & Kim, 2024). More recently, PLS-SEM research has confirmed the predictive role of self-regulation in GenAI-assisted writing (Jin et al., 2025). Given these findings, it is reasonable to expect that self-regulated EFL students will actively manage their interactions with GenAI. This leads to the first hypothesis (H1).

H1: Self-regulation can positively affect EFL students' effective use of GenAI in writing.

4.2 Critical thinking

Critical thinking is commonly defined as the capacity for reflective and purposeful judgment that enables individuals to analyze, evaluate, and synthesize information for informed decision-making (Facione, 1990; Halpern, 2013). Within the SCT framework, critical thinking functions as a personal cognitive resource that supports behavioral regulation. It allows learners to monitor, evaluate, and adjust their engagement with external inputs, serving as a key mechanism through which agency is enacted in learning (Bandura, 2020; Zimmerman, 2002). In this study, critical thinking refers to EFL students' ability to critically assess GenAI-generated content by judging its relevance and accuracy, integrating useful insights, and discarding those that do not contribute to writing quality.

Previous research has demonstrated a strong link between critical thinking and self-regulation. Zimmerman (2002) argued that critical thinking underpins the monitoring and adjustment processes essential to self-regulated learning, while Elder and Paul (2004) emphasized its role in the continual refinement of thought. Further educational studies highlight that critical thinking is crucial for effective problem-solving and decision-making in technology-supported environments (Snyder & Snyder, 2008). Empirical research also showed that students with well-developed critical thinking skills demonstrate better performance in writing, revision, and editing tasks (Carroll, 2007; Mehta & Al-Mahrooqi, 2015). These findings suggest that EFL learn-

ers who think critically are more likely to engage with GenAI outputs in a discerning and constructive manner. Therefore, this study proposes the following hypotheses.

H2: Critical thinking can positively affect EFL students' effective use of GenAI in writing.

H3: Critical thinking can positively affect EFL students' self-regulation.

H4: Critical thinking can positively affect EFL students' GenAI literacy.

4.3 Growth mindset

Growth mindset refers to the belief that one's abilities, including cognitive and academic skills, can be developed through sustained effort and practice (Dweck, 2006). Within the SCT framework, it represents a key personal motivational resource that shapes how learners approach challenges and regulate their behavior over time (Schunk & DiBenedetto, 2020). In this study, growth mindset is defined as EFL students' belief that their writing ability can improve through continuous effort and engagement.

Empirical research in EFL contexts has shown that growth mindset supports self-regulated learning by motivating students to set goals, maintain effort, and adjust strategies when facing difficulties (Xu & Wang, 2022). It has also been linked to enhanced critical thinking, as learners who adopt this belief tend to view challenges as opportunities for development (Tu, 2024). Based on these insights, we hypothesize that in GenAI-assisted writing, such learners are more likely to treat GenAI as a learning resource rather than a shortcut. Consequently, we propose three hypotheses listed below.

H5: Growth mindset can positively affect EFL students' effective use of GenAI in writing.

H6: Growth mindset can positively affect EFL students' self-regulation.

H7: Growth mindset can positively affect critical thinking.

4.4 General writing competence

General writing competence refers to an individual's ability to convey meaning effectively through written language, including accurate vocabulary use, grammatical control, and coherent organization (Weigle, 2007). Within the SCT framework, general writing competence is considered part of learners' personal attributes. More specifically, it reflects their domain-specific knowledge and skills that support agentic functioning by enabling purposeful and context-sensitive written expression (Bandura, 2020). In this study, general writing competence is defined as EFL students' ability to produce coherent and well-structured English texts, which reflect core features of agentic learning behavior. Drawing on prior work (Gabrielatos, 2002; Hyland, 2003; Weigle, 2007), this construct is conceptualized across four dimensions: (1) language use, assessing grammatical accuracy, vocabulary control, and clarity of expression; (2) organization and layout, focusing on logical flow and structural coherence; (3) detail and specificity, emphasizing the inclusion of concrete examples and precise

elaboration; and (4) task relevance, evaluating alignment with assignment requirements and reader expectations.

Empirical evidence has indicated that learners with higher writing competence tend to perform better in revision, idea development, and overall task fulfillment (Eley, 1953; Yasuda, 2011). Thus, even in GenAI-assisted writing, general writing competence might remain a key determinant of effective performance. In addition, writing competence has demonstrated positive associations with both growth mindset and critical thinking. Students with stronger writing abilities are more likely to persist through challenges and make constructive use of feedback (Cheong et al., 2023; Yao et al., 2024), and are also better equipped to evaluate, synthesize, and integrate information effectively (Nazarov & Kuvandikova, 2023; Sánchez-Peña & Chapetón, 2018). Based on these insights, three research hypotheses are proposed.

H8: General writing competence can positively affect EFL students' effective use of GenAI in writing.

H9: General writing competence can positively affect EFL students' growth mindset.

H10: General writing competence can positively affect EFL students' critical thinking.

4.5 GenAI literacy

GenAI literacy goes beyond basic operational skills to include a multidimensional set of capabilities essential for effective GenAI engagement (Zhao et al., 2024). It comprises four core dimensions: conceptual knowledge (understanding how large language models generate text and their limitations), practical skills (designing effective prompts, strategic interaction, and navigating outputs), evaluative judgment (critically assessing and integrating GenAI content), and ethical awareness (responsible use regarding authorship, integrity, and bias) (Ridley & Pawlick-Potts, 2021).

In this study, GenAI literacy is defined as EFL students' informed and strategic use of GenAI across all stages of English writing, including prompt generation, feedback evaluation, content revision, and ethical reflection (Feng Teng, 2024). Within the SCT framework, GenAI literacy reflects students' capacity to proactively engage with environmental affordances and coordinate their behavioral regulation in GenAI-assisted EFL writing (Bandura, 2020). Rather than passively accepting GenAI outputs, students with high GenAI literacy transform these tools into meaningful resources to achieve their writing goals, supporting ongoing monitoring, reflective revision, and sustained self-regulation throughout the writing process.

Existing research has underscored the growing importance of GenAI literacy in educational contexts (Kim et al., 2025; Lee, 2023). In particular, it has been associated with self-directed learning and improved task outcomes in language education (Sullivan et al., 2024). Given these findings, it is reasonable to hypothesize that GenAI literacy supports more effective use of GenAI tools in EFL writing. Therefore, this study postulates the eleventh hypothesis:

H11: GenAI literacy can positively affect EFL students' effective use of GenAI in writing.

5 Methodology

5.1 Participants

A total of 356 EFL learners from a Chinese Double First-Class university (i.e., world-class universities and disciplines) volunteered for this study. This university was selected for two key reasons: first, it is an early adopter of integrating GenAI into language education, having piloted GenAI use in college English courses since 2022. Second, the university's EFL learners, being digital natives, frequently use GenAI, with an average usage score of 4.30 out of 5 for general tasks and 4.17 for writing tasks. This high usage frequency ensured their active participation in GenAI-assisted writing activities.

Participants were first-year college students enrolled in a required English course designed to improve overall language skills with an emphasis on writing. To enhance writing, teachers integrated GenAI tools (e.g., ChatGPT) and provided guidelines for using these tools at various writing stages—such as for real-time feedback on linguistic accuracy, style, and idea generation. Participation in the survey was voluntary, with no incentives or penalties. Fourteen incomplete questionnaires were excluded, resulting in 342 valid responses. Among these 342 participants, 42.98% were male ($n=147$) and 57.02% were female ($n=195$). Table 1 summarizes the demographic characteristics.

5.2 Instruments

This study used a combination of a questionnaire and writing tasks to examine the hypotheses within the proposed research model.

5.2.1 Questionnaire

To measure self-regulation, critical thinking, growth mindset, and GenAI literacy, a two-part questionnaire was developed. The first part gathered demographic information such as age and gender. The second part contained 16 items—four for each con-

Table 1 Demographic characteristics of the participants

	N	%
Gender		
Male	147	42.98
Female	195	57.02
Total	342	100
Age		
17	179	52.34
18	163	47.66
Total	342	100

struct–adapted from established instruments (Dweck, 2006; Halpern, 2013; Kaplan et al., 2019; Lee, 2023; Mehta & Al-Mahrooqi, 2015; Zhao et al., 2024) and refined by three experienced EFL writing instructors to fit GenAI-assisted writing contexts. Table 2 presents the specific items and references. All items were rated on a seven-point Likert scale (1 = strongly disagree, 7 = strongly agree).

The questionnaire was carefully developed and validated. Initially, it was translated into simplified Chinese and piloted with ten English majors. Based on their feedback, minor adjustments were made for clarity. The revised version was then back-translated into English by a certified translation instructor to ensure semantic accuracy. Reliability and validity were evaluated using data from 342 valid responses. As shown in Table 3, Cronbach’s alpha ranged from 0.811 to 0.894, all above the recommended 0.800 threshold. Composite reliability (CR) ranged from 0.888 to 0.926, and Average Variance Extracted (AVE) values exceeded 0.70 for all constructs, supporting strong internal consistency and convergent validity. All outer loadings were 0.780 or higher, and HTMT ratios remained below 0.850, indicating

Table 2 Measurement items

Construct	Description of items	References
Self-regulation	SR1: I set specific goals for using GenAI, such as improving language accuracy or enhancing clarity in my English writing (Goal setting) SR2: I identify which parts of my English writing would benefit the most from GenAI feedback (Planning) SR3: While using GenAI, I check if I understand the contents generated and make sure they fit what I want to express (Monitoring use) SR4: I evaluate how effectively I use GenAI to enhance my English writing, including language quality and content coherence (Reflecting)	Kaplan et al. (2019) Zimmerman and Kitsantas (2014)
Critical thinking	CT1: I check GenAI-generated outputs for errors, inconsistencies, or irrelevant information (Evaluating GenAI outputs) CT2: I evaluate the reliability and relevance of GenAI-generated outputs before using them (Assessing credibility) CT3: I engage in multiple rounds of interaction with GenAI to refine the outputs, critically considering each suggestion (Refining through interaction) CT4: I decide which GenAI feedback will best improve clarity, coherence, and alignment with my writing goals. (Choosing effective strategies)	Carroll (2007) Halpern (2013) Mehta and Al-Mahrooqi (2015)
Growth mindset	GM1: I can always improve my English writing competence GM2: No matter who I am, I can significantly improve my English writing competence GM3: No matter how much intelligence I have for English writing, there’s always room for improvement GM4: In learning English writing, if I work hard, I will always get better	Dweck and Leggett (1988) Dweck (2006)
GenAI literacy	GL1: I understand the basic principles and how GenAI works in the context of English writing (Understanding GenAI) GL2: I know how to prompt and interact with GenAI for researching, organizing, and communicating information effectively (Effective prompting and interaction) GL3: I can use prompts and interact with GenAI to solve specific problems in my English writing (Problem-solving with GenAI) GL4: I am aware of the ethical and legal considerations involved in using GenAI tools for writing (Ethical awareness)	Lee (2023) Ridley and Pawlick-Potts (2021) Zhao et al. (2024)

Table 3 Reliability and convergent validity results

Constructs	Items	Outer Loadings	Cronbach's alpha	CR	AVE
Self-regulation	SR1	0.864	0.867	0.909	0.715
	SR2	0.889			
	SR3	0.844			
	SR4	0.780			
Critical thinking	CT1	0.842	0.886	0.922	0.746
	CT2	0.868			
	CT3	0.855			
	CT4	0.890			
Growth mindset	GM1	0.841	0.870	0.911	0.720
	GM2	0.827			
	GM3	0.855			
	GM4	0.869			
General writing competence	GWC1	0.875	0.894	0.926	0.759
	GWC2	0.832			
	GWC3	0.897			
	GWC4	0.881			
GenAI literacy	GL1	0.843	0.869	0.911	0.718
	GL2	0.861			
	GL3	0.854			
	GL4	0.832			
Effective use of GenAI in writing	EGW1	0.851	0.811	0.888	0.725
	EGW2	0.863			
	EGW3	0.840			

good item reliability and discriminant validity. Together, these results confirm that the questionnaire reliably measures the intended constructs in the context of GenAI-assisted EFL writing.

5.2.2 Writing tasks

To evaluate general writing competence and the effective use of GenAI in writing, participants completed English writing tasks under two conditions: one without GenAI assistance and one with it. Across the two conditions, participants wrote two cause-and-effect essays—one analyzing the causes of a phenomenon and the other examining its effects. The task without GenAI was used to assess general writing competence across four dimensions: language, organization and layout, detail and specificity, and task relevance (Gabrielatos, 2002; Hyland, 2003; Weigle, 2007). Two experienced EFL writing teachers rated these dimensions using a seven-point rating scale.

In the GenAI-assisted condition, participants completed a comparable essay using GenAI tools, with their writing sessions recorded via EV Capture for further analysis. The effectiveness of GenAI use was evaluated across three dimensions: GenAI support, interaction quality, and final quality, drawing on relevant frameworks from the literature (Barrot, 2023; Guo & Wang, 2024; Mizumoto & Eguchi, 2023). GenAI support measured how well participants used GenAI to improve language accuracy

and content quality. Interaction quality assessed their ability to critically evaluate and integrate feedback, while final quality focused on the overall quality of the final text. These dimensions were scored on a 100-point scale to capture the dynamic and multifaceted nature of GenAI-assisted writing more precisely. Unlike general writing competence, which involves distinct skills that can be systematically assessed using an analytic rating scale, GenAI-assisted writing is more dynamic, involving real-time problem-solving and content enhancement. A 100-point scale allowed for a more detailed evaluation of participants' use of GenAI.

5.3 Data collection procedure

This study was conducted during the first semester of the 2024–2025 academic year with first-year college students enrolled in a required English course. Throughout the semester, teachers guided students in using GenAI at various stages of the writing process, so that by semester's end, students had substantial experience with these tools.

Data were collected in three stages during the final phase of the semester. In the first stage, conducted before the last week of class, students completed a growth mindset scale to assess their motivational beliefs about writing. In the second stage, held during the final week, participants completed two cause-and-effect writing tasks of approximately 300 words each. One was completed without access to GenAI tools and served as a baseline measure of general writing competence. The other was completed with the aid of ChatGPT 3.5, during which students were encouraged to use the tool freely throughout the writing process. Their interactions were recorded using EV Capture to document actual GenAI engagement. This task was used to assess their effective use of GenAI in writing. To control for potential order effects, task sequence was counterbalanced: some students completed the non-GenAI task first, while others began with the GenAI-assisted task. In the final stage, immediately after the writing tasks, students completed post-task surveys measuring self-regulation, critical thinking, and GenAI literacy. This post-task assessment allowed students to reflect on their experiences throughout the semester, providing well-rounded insights into the above constructs regarding their engagement with GenAI in writing.

5.4 Data analysis

Data analysis was conducted in several stages to ensure methodological rigor and transparency. First, questionnaire responses were screened using SPSS 22. Cases with substantial missing data or patterned responses were excluded. Univariate normality was assessed via skewness and kurtosis, and multivariate outliers were screened using Mahalanobis distance. The final dataset comprised 342 valid responses. Descriptive statistics were calculated to profile the sample and examine variable distributions. Writing performance was evaluated by two experienced English writing instructors who independently rated each student's two tasks after agreeing on rubric interpretations. The non-GenAI-assisted task was scored on four dimensions—language use, organization and layout, detail and specificity, and task relevance—using 7-point rating scales, based on established criteria (Gabrielatos, 2002; Hyland, 2003;

Weigle, 2007). These scores were averaged to form a composite measure of general writing competence (GWC), with good internal consistency ($\alpha=0.894$). The GenAI-assisted task was rated on a 100-point rubric with three dimensions (40/30/30): extent of GenAI support, interaction quality, and final output quality (Barrot, 2023; Guo & Wang, 2024; Mizumoto & Eguchi, 2023). The sum of these ratings formed the effective GenAI use composite, supported by high reliability ($\alpha=0.811$). Inter-rater agreement exceeded 0.85 across all dimensions, and discrepancies were resolved via discussion with a third expert.

Importantly, the GenAI-assisted and non-GenAI writing tasks were not analyzed as two separate groups using traditional between-group comparisons. Instead, both tasks were completed by every participant within a within-subjects design, resulting in two writing scores per participant: one for the non-GenAI task (general writing competence) and one for the GenAI-assisted task (effective GenAI use). In the PLS-SEM analysis, these two scores were modeled as distinct continuous variables, each representing a separate latent construct within the structural model. Specifically, the non-GenAI score served as the “general writing competence” latent variable, while the GenAI-assisted score served as the “effective GenAI use” latent variable. Both, together with other questionnaire-based constructs (self-regulation, critical thinking, growth mindset, GenAI literacy), were incorporated into a single structural model to examine their direct and indirect relationships. Rather than dividing participants into groups, the model estimated the structural paths among all latent variables for the entire sample. This approach allowed us to directly assess, for each participant, how baseline writing competence and individual characteristics contributed to GenAI-assisted writing performance.

To ensure consistency across variables measured on different scales, all continuous indicators were standardized before model estimation. Specifically, all item-level scores (questionnaire-based item scores and the weighted dimension scores for the GenAI use measures) were transformed into z-scores using the formula $z=(x-\mu)/\sigma$, where μ represents the sample mean and σ the standard deviation. All standardization was conducted using SPSS 22 prior to inclusion in the PLS-SEM analysis.

Finally, PLS-SEM analysis was then conducted using SmartPLS 3.0, following a two-step approach: validation of the measurement model (reliability, convergent validity) and testing of the structural model to examine hypothesized relationships.

6 Results

6.1 Descriptive analysis of key personal factors (RQ1)

Table 4 presents the descriptive statistics for each construct. All skewness and kurtosis scores fell within the recommended normality range as suggested by Hair et al. (2006).

The mean values for the first five constructs ranged from 4.583 (SD=1.212) to 4.947 (SD=1.276), all scoring above average but below 5. GenAI literacy had the highest rating ($M=4.947$, $SD=1.276$), followed by critical thinking ($M=4.755$, $SD=1.351$), self-regulation ($M=4.650$, $SD=1.289$), and general writing compe-

Table 4 Descriptive analysis of each construct

Constructs	Means	SD	Skewness	Kurtosis
Self-regulation (SR)	4.650	1.289	−0.444	−0.687
Critical thinking (CT)	4.755	1.351	−0.526	−0.467
Growth mindset (GM)	4.583	1.212	−0.571	−0.226
General writing competence (GWC)	4.621	1.280	−0.449	−0.607
GenAI literacy (GL)	4.947	1.276	−0.731	−0.035
Effective use of GenAI in writing (EGW)	79.614	7.246	−0.339	−1.417

Table 5 Heterotrait-Monotrait Ratio (HTMT) results

	CT	EGW	GL	GWC	GM	SR
CT						
EGW	0.575					
GL	0.379	0.576				
GWC	0.462	0.543	0.51			
GM	0.303	0.373	0.363	0.379		
SR	0.422	0.574	0.484	0.376	0.409	

tence ($M=4.621$, $SD=1.280$). These findings suggest that EFL students felt comfortable using GenAI tools and had basic writing skills, and their critical thinking and self-regulation in GenAI-assisted writing were modest. Growth mindset had the lowest mean ($M=4.583$, $SD=1.212$), indicating a moderate belief in their potential to improve writing through effort. The comparable ratings across writing abilities, motivational, cognitive, and technological constructs likely illustrate these factors were equally important in the effective use of GenAI in writing.

Besides, effective use of GenAI in writing had a mean of 79.614 ($SD=7.246$). While this score reflects a fair level of GenAI use, it also reveals the room for further improvement.

6.2 PLS-SEM model assessment for structural relations (RQ2)

The PLS-SEM model assessment consisted of two phases. First, we analyzed the measurement model to evaluate the reliability and validity of the observable indicators. Next, we assessed the structural model to test the overall model fit and examine the hypothesized relationships.

6.2.1 Analysis of the measurement model

The measurement model was assessed for reliability, convergent validity, and discriminant validity (Hair et al., 2013). Reliability was evaluated using Cronbach's alpha, outer loadings, and composite reliability (CR). Convergent validity was examined through average variance extracted (AVE), while discriminant validity was tested using the Heterotrait-Monotrait ratio (HTMT) (Henseler et al., 2015). The results are shown in Tables 3 and 5.

As shown in Table 3, all items had outer loadings above 0.700, indicating good indicator reliability. Cronbach's alpha values ranged from 0.811 to 0.894, exceeding the recommended minimum of 0.700, confirming satisfactory internal consistency. Moreover, CR values for all constructs were above the threshold of 0.700, further affirming the reliability of the measures (Kline, 2005). All AVE values surpassed 0.500, ranging from 0.715 to 0.759, which meets the criteria for convergent validity (Fornell & Larcker, 1981). These results suggest that the items effectively represent their respective constructs.

As presented in Table 5, all HTMT values were below the recommended cutoff of 0.850, confirming adequate discriminant validity between the constructs (Henseler et al., 2015). This indicates that each construct in the model is sufficiently distinct from the others. Overall, these results validate the measurement model as both reliable and valid.

6.2.2 Analysis of structural model

Before conducting further analyses, it is essential to verify the model fit and predictive ability. The Goodness of Fit (GOF) index was used to evaluate the PLS-SEM model fit, along with R^2 values and the Stone-Geisser Q^2 test to examine predictive performance. The results are presented in Table 6.

The GOF index ranges from 0 to 1, with values of 0.100, 0.250, and 0.360 indicating small, moderate, and large model fit, respectively (Wetzels et al., 2009). In this study, the calculated GOF value was 0.393, which exceeds the threshold for a large effect size, indicating good over all model fit. R^2 values show how well the model predicts each construct. Effective use of GenAI in writing had the highest R^2 value (0.432), meaning that 43.2% of its variance is explained by the model, indicating strong predictive power (Chin, 1998). Other constructs, such as critical thinking ($R^2=0.189$), self-regulation ($R^2=0.212$), and growth mindset ($R^2=0.113$), showed lower R^2 values, signifying moderate to weak predictive power. The Q^2 values further support the model's predictive relevance, with all endogenous latent variables having values greater than zero (Hair et al., 2013).

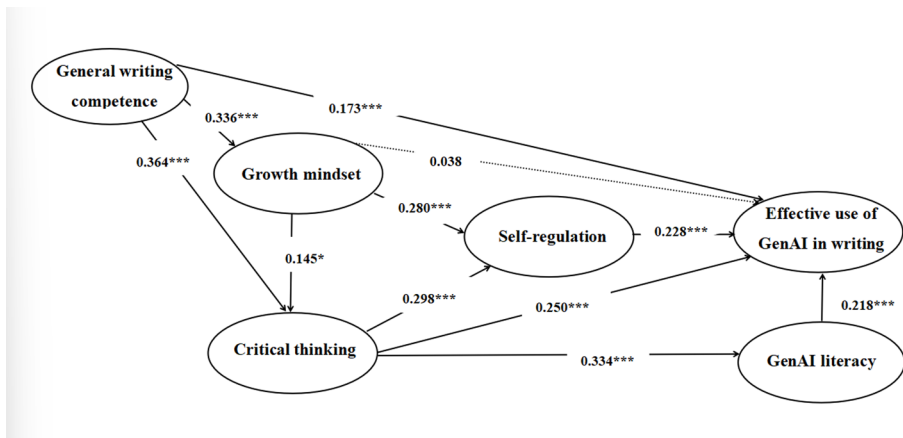
After confirming the model fit, we proceeded with the structural model analysis to examine the hypothesized relationships. Path coefficients (β) and corresponding p-values were evaluated using a bootstrapping procedure with 5,000 resamples (Hair et al., 2013). The results of the structural paths are presented in Table 7 and illustrated in Fig. 2.

Table 6 Model fit of proposed model

Construct	AVE	R^2	Q^2
Critical thinking	0.746	0.189	0.137
GenAI literacy	0.718	0.112	0.078
General writing competence	0.759	/	/
Growth mindset	0.720	0.113	0.079
Self-regulation	0.715	0.212	0.147
Effective use of GenAI in writing	0.725	0.432	0.302
GOF = $\sqrt{\text{AVE} \times R^2}$	0.393		

Table 7 Path coefficients and hypotheses testing

Causal path	Path coefficient	Standard deviation	<i>T</i> statistics	<i>P</i> values	Supported
H1: SR→EGW	0.228	0.048	4.759	0.000	yes
H2: CT→EGW	0.250	0.046	5.411	0.000	yes
H3: CT→SR	0.298	0.050	5.927	0.000	yes
H4: CT→GL	0.334	0.051	6.612	0.000	yes
H5: GM→EGW	0.038	0.045	0.832	0.405	no
H6: GM→SR	0.280	0.054	5.229	0.000	yes
H7: GM→CT	0.145	0.058	2.510	0.012	yes
H8: GWC→EGW	0.173	0.048	3.647	0.000	yes
H9: GWC→GM	0.336	0.058	5.805	0.000	yes
H10: GWC→CT	0.364	0.053	6.855	0.000	yes
H11: GL→EGW	0.218	0.049	4.462	0.000	yes

**Fig. 2** PLS-SEM model for EFL students' effective use of GenAI in writing

As displayed, 10 out of the 11 hypothesized paths were supported. Four individual factors—self-regulation, critical thinking, GenAI literacy, and general writing competence—were significantly related to the effective use of GenAI in writing, confirming H1, H2, H8, and H11. Among these, critical thinking ($\beta=0.250$, $p<0.001$) and self-regulation ($\beta=0.228$, $p<0.001$) were the strongest predictors, followed by GenAI literacy ($\beta=0.218$, $p<0.001$). In contrast, general writing competence had a lower yet significant effect ($\beta=0.173$, $p<0.001$).

Surprisingly, growth mindset did not have a direct significant impact on the effective use of GenAI ($\beta=0.038$, $p=0.405$), rejecting H5. However, growth mindset significantly influenced self-regulation ($\beta=0.280$, $p<0.001$) and critical thinking ($\beta=0.145$, $p=0.012$) (supporting H6 and H7), suggesting an indirect influence on GenAI use through these factors. Additionally, general writing competence was found to significantly predict growth mindset ($\beta=0.336$, $p<0.001$) and critical thinking ($\beta=0.364$, $p<0.001$), confirming H9 and H10. Critical thinking also emerged as a key determinant of self-regulation ($\beta=0.298$, $p<0.001$) and GenAI literacy ($\beta=0.334$, $p<0.001$), supporting H3 and H4. Overall, the findings highlight the com-

plex interactions among motivational, cognitive, and technological factors and how they collaboratively shape EFL students' engagement with GenAI in writing.

7 Discussion

This study proposed and empirically tested a research model for EFL students' effective use of GenAI in writing, finding support for 10 out of the 11 hypotheses.

The results reveal that self-regulation, critical thinking, GenAI literacy, and general writing competence all significantly contributed to the effective use of GenAI in writing. Among these, critical thinking and self-regulation were the strongest predictors, highlighting the importance of higher-order cognitive skills in managing GenAI-assisted tasks. These findings are consistent with prior studies highlighting the value of such abilities in planning, evaluating, and revising digital work while maintaining authorship and agency (Guo & Lee, 2023; Jin et al., 2025; Li & Kim, 2024). Therefore, our results indicate that EFL students with stronger self-regulation managed their writing processes more strategically, while those with higher critical thinking evaluated AI outputs effectively and made informed decisions about their use.

Notably, our results extend the literature by revealing a directional relationship: critical thinking significantly predicted self-regulation. Students with strong critical thinking skills are better able to monitor and adapt their writing processes when working with GenAI. In practice, these students routinely engage in evaluative questioning when assessing GenAI-generated contents—for example, asking themselves whether a sentence is logically consistent with their intended meaning, whether a suggestion enhances clarity or introduces ambiguity, or whether GenAI-generated evidence is factually accurate. Through such critical evaluation, they can detect inaccuracies, logical inconsistencies, or stylistic mismatches in GenAI outputs. Based on these judgments, students activate self-regulatory strategies such as rephrasing for precision (monitoring and revision), restructuring for coherence (goal-setting and strategic planning), cross-checking outputs with external sources (monitoring and evaluative judgment), and experimenting with alternative prompts to improve results (strategic adaptation and self-directed revision). Collectively, these behaviors illustrate how critical thinking may activate specific phases of self-regulation, reflecting a meta-cognitive process aligned with Zimmerman's (2002) framework, in which evaluative judgment informs goal-setting, monitoring, and adaptive revision. This interaction also exemplifies the SCT triadic model, in which cognitive appraisals (personal factors) shape behavioral responses to GenAI feedback (environmental affordances). The reciprocal nature of this process suggests that fostering critical thinking can strengthen students' self-regulatory engagement in GenAI-assisted writing. Overall, these findings advance our understanding by clarifying the cognitive mechanisms linking critical evaluation and strategic regulation, and by situating these interactions within a theoretical framework that emphasizes learner agency.

The results further indicate that, beyond cognitive factors, GenAI literacy is a strong predictor of effective GenAI use in writing. This result supports prior research emphasizing the importance of technological knowledge and skills for optimizing

GenAI use (Kim et al., 2025; Lee, 2023; Sullivan et al., 2024; Zhao et al., 2024). Both the operational proficiency and the deeper understanding of GenAI tools play key roles in how students engage with GenAI across various natural language tasks, including EFL writing. Moreover, our results further reveal that critical thinking significantly predicts GenAI literacy, suggesting that students' ability to contextualize and strategically apply GenAI tools is shaped not merely by technological know-how but also by higher-order cognitive abilities. In other words, while technological literacy is crucial, it is insufficient on its own for maximizing GenAI's benefits in writing. Cognitive skills—especially critical thinking—determine how effectively EFL students develop and apply their GenAI literacy. This relationship can also be interpreted through the triadic model of SCT, in which cognitive attributes mediate interactions between behavior and environment. In this case, critical thinking enables learners to transform their exposure to GenAI into deeper literacy, reinforcing the view that cognitive scaffolding is foundational to effective technology integration in EFL writing. These findings reinforce the primacy of cognitive factors in mediating students' engagement with GenAI.

Interestingly, general writing competence exerted a significant but weaker direct effect on effective GenAI use compared to cognitive and technological factors. This contrasts with earlier findings in traditional writing contexts, where writing competence was often identified as the most decisive factor (Eley, 1953; Yasuda, 2011). One possible explanation is that the powerful generative capabilities of GenAI reduce EFL students' reliance on conventional writing skills. For instance, GenAI enhances language accuracy through grammar corrections and lexical suggestions, supports organization by structuring ideas, and provides concrete examples, enabling students to produce coherent texts with limited linguistic effort. Furthermore, GenAI enables content adaptation, thereby mitigating the impact of baseline writing competence on performance.

Notably, our study also reveals that general writing competence significantly predicts critical thinking. In other words, EFL students with stronger writing skills are better able to engage in critical analysis, which in turn enhances their decision-making and evaluation when interacting with GenAI outputs. From a SCT perspective (Bandura, 1986), writing competence functions as a personal knowledge resource that shapes behavioral engagement and learning outcomes within a technology-rich environment. Therefore, while the direct impact of writing competence may appear reduced in GenAI-assisted contexts, its indirect influence remains critical. These results call for a reconceptualization of writing competence in the AI era, not merely as a linguistic skillset, but as a scaffold for higher-order cognitive regulation that underpins effective technology integration in academic writing.

Surprisingly, growth mindset did not show a direct significant impact on the effective use of GenAI in writing, which contradicts previous research that identified motivation as a key driver of writing performance (Xu & Wang, 2022; Yao et al., 2024). One plausible explanation is that the immediacy and generative power of GenAI tools may diminish the need for sustained personal effort. When high-quality content can be produced with minimal effort, students may bypass iterative drafting and reflection, reducing the observable impact of motivational constructs such as growth mindset.

Despite the absence of a direct effect, our findings show that growth mindset significantly predicted both critical thinking and self-regulation, highlighting an important indirect pathway for GenAI-assisted writing. EFL students who believe their abilities are malleable are more likely to engage in meta-cognitive reflection and strategic revision when using GenAI. This process typically begins with a learning-oriented attitude: students with a growth mindset view GenAI outputs as provisional drafts rather than definitive answers. When encountering GenAI-generated suggestions, they are inclined to pause and critically reflect, for example, by asking themselves whether a particular sentence accurately captures their intended meaning, whether the argument is logically valid, or whether the phrasing can be improved for clarity. This mindset motivates them to revise, experiment with alternative phrasing, and restructure their work as needed, seeing revision as an integral part of learning rather than as a sign of failure. Through this reflective approach, growth mindset fosters a cycle of active monitoring, iterative revision, and exploration of alternatives when GenAI outputs fall short. Such behaviors constitute a meta-cognitive loop: students evaluate GenAI input, generate hypotheses for improvement, implement changes, and reassess the results. The belief that “I can improve this” sustains their willingness to test revisions and restructure paragraphs without fear of making mistakes. Over time, such repeated cycles of reflection and revision not only strengthen self-regulation skills but also reinforce students’ sense of agency and their perception of GenAI as a valuable learning partner. This pattern illustrates how motivational beliefs (personal factors) are translated into cognitive and behavioral engagement with GenAI (environmental affordances), aligning with the triadic model of SCT. Furthermore, the reciprocal prediction between general writing competence and growth mindset suggests a recursive cycle: students who believe they can improve are more likely to self-monitor and revise, which reinforces both their sense of control and the utility of GenAI. Taken together, these findings emphasize that growth mindset should be viewed not in isolation, but as part of an interconnected system of motivational, cognitive, and linguistic factors that support self-directed learning with GenAI.

The unexpected lack of a direct effect invites further exploration. It is possible that additional psychological or sociological variables, such as technology self-efficacy, facilitating conditions, teacher support, or awareness of GenAI ethics, may moderate the relationship between growth mindset and effective GenAI use. Future research should incorporate these factors to refine the model and clarify the conditional pathways through which growth mindsets support GenAI-assisted writing.

In conclusion, EFL students’ effective use of GenAI tools in writing is primarily driven by cognitive factors like self-regulation and critical thinking, while GenAI literacy also plays a key supportive role. Unlike previous beliefs that highlighted general writing competence and motivation as central, these factors now seem to play a less significant or indirect role by reinforcing cognitive engagement in the GenAI-assisted environment. Therefore, future teaching should focus on developing EFL students’ cognitive skills, fostering adaptive motivation, developing integrated GenAI literacy, and rethinking the concept of writing competence in the age of GenAI.

8 Implications

8.1 Theoretical implications

The findings of this study offer several important theoretical contributions. Drawing on SCT, our model integrates cognitive factors (self-regulation, critical thinking), technological competence (GenAI literacy), motivational orientation (growth mindset), and general writing competence to provide a holistic perspective on how these individual factors interact to shape EFL students' effective use of GenAI in writing. Grounded in the triadic reciprocity framework, the findings underscore how personal attributes influence behavioral engagement with GenAI tools, reinforcing the SCT principle that learner agency is central to navigating technology-enhanced learning environments.

Notably, while cognitive and technological factors directly influence GenAI use, growth mindset exerts its impact indirectly by strengthening students' critical thinking and self-regulation. This suggests that future models might benefit from repositioning growth mindset as a second-order predictor that operates primarily through its influence on cognitive mediators. Future research should further refine the model by examining additional psychological and sociocultural moderators, such as technology self-efficacy, teacher support, and attitudes toward GenAI.

Moreover, our results contribute to the broader discussion in educational technology by highlighting a shift in the determinants of writing effectiveness in the GenAI era. Traditional perspectives that emphasize general writing competence and motivation may need to be updated to reflect the growing importance of cognitive and technological skills in GenAI-supported environments. The integration of these dimensions suggests that effective writing in the GenAI era relies on a dynamic interplay among motivational, cognitive, and technological factors.

8.2 Practical implications

Several pedagogical implications can also be drawn to enhance EFL writing instruction in the GenAI era. These suggestions focus on fostering cognitive skills, integrating GenAI literacy, refining traditional writing competence, effectively motivating students within GenAI-assisted environments, and addressing ethical concerns.

Given the dominant role of cognitive skills in EFL students' effective use of GenAI, instructional design should prioritize cultivating self-regulation and critical thinking through structured, process-oriented activities. To nurture self-regulation, classroom activities can begin with explicit goal-setting exercises. Teachers might introduce pre-task planning sheets in which students articulate their objectives for GenAI use, such as clarifying idea flow, improving coherence, or expanding supporting details. Before drafting, students complete these sheets to identify target areas for revision and anticipate how GenAI might help. During writing, teachers can provide monitoring checklists for students to use as they interact with GenAI. For every GenAI suggestion, students indicate whether they accept, modify, or reject it, and briefly explain their reasoning. These notes can be annotated in the draft or summarized in a process log. After completing the writing task, students engage in reflective writing logs to

review their GenAI interactions. In these logs, they summarize which suggestions were most helpful, which were revised or dismissed, and how their choices related to their initial goals. Teachers can prompt reflection with questions such as: “How did GenAI help you meet your writing objectives?”; “Which suggestions did you reject and why?”; and “How did your own revisions improve the text?” This iterative process (goal setting, progress tracking, and reflective evaluation) reinforces authorship and reduces passive reliance on GenAI-generated content.

To cultivate critical thinking, instructional interventions should explicitly train students to assess the quality and appropriateness of GenAI outputs. Teachers can design evaluative tasks where students systematically review GenAI-generated paragraphs for coherence, factual accuracy, and stylistic alignment with academic discourse. For instance, students may use highlighting tools to mark logical inconsistencies, annotate unclear reasoning, or suggest alternative phrasings directly on the text. These activities can be scaffolded with analytic rubrics or targeted checklists to support students’ evaluations. Another effective classroom activity is comparative drafting: students produce one draft independently and another with GenAI assistance. They then perform a side-by-side analysis to determine which version better achieves clarity, coherence, or rhetorical purpose, and discuss the specific revisions needed in the GenAI-assisted draft to meet academic standards. Such practices illuminate the cognitive work involved in meaningfully adapting GenAI outputs, fostering deeper awareness of audience, context, and communicative intent.

Considering the supportive role of GenAI literacy in facilitating effective writing, instruction should move beyond functional training toward cultivating students’ ability to apply GenAI tools critically and ethically. To achieve this, teachers can scaffold GenAI literacy development across three progressive stages: conceptual grounding, procedural fluency, and critical and ethical application. At the conceptual stage, instruction should focus on building students’ foundational understanding of GenAI: what it is, how it generates language, and its specific affordances and limitations for EFL writing. This can be conducted through interactive lectures or short modules covering key concepts such as machine learning, training data, and the typical features of GenAI tools. Teachers might incorporate live demonstrations by showing how GenAI responds to different prompts in real time, followed by class discussion to highlight both strengths and pitfalls such as hallucinations or biases. At the procedural stage, students move beyond interface navigation to explore how prompt specificity, tone, and structure influence output. Teachers might design structured prompt engineering workshops. For example, students could be assigned a simple writing task (“Describe your favorite place”) and experiment with altering prompt wording, tone, length, or audience. Class discussion can then focus on how these changes influence GenAI output. Assignments such as a “prompt portfolio,” where students record prompts, results, and analyses of what works best, help them gain practical control and awareness of the tool’s responsiveness. At the evaluative and ethical stage, instruction targets critical and responsible GenAI use. Teachers can assign peer review exercises, using rubrics to evaluate the coherence, factual accuracy, and audience appropriateness of GenAI-generated drafts. Questions such as “Does the paragraph logically support the thesis?” or “Is the evidence accurate and appropriately cited?” can guide critical evaluation. Comparative drafting assign-

ments can be used, where students write one version of a paragraph independently and another with GenAI, then reflect in writing on what GenAI improved, what errors or biases were introduced, and how their own revisions addressed these. Classroom debates or case studies about responsible AI use—such as when GenAI use crosses ethical boundaries or how to acknowledge AI-generated content—can further promote ethical reflection. Taken together, this instructional model enables students to progress from basic tool use to intentional, reflective, and ethical engagement with GenAI in EFL writing. With sustained guidance, students come to see GenAI not as a shortcut, but as a dialogic partner—one that supports, but never replaces, their own voice and agency.

General writing competence had a lower effect compared to cognitive and technological factors, indicating that traditional writing competence must evolve in the GenAI-assisted learning environment. Rather than being treated as a stand-alone skill, writing competence should be integrated with technological awareness and active human-AI collaboration. For instance, to improve language accuracy, teachers can design activities where students use GenAI tools to receive immediate grammar corrections and word choice suggestions. Following this, students might engage in a structured reflection session in which they review the corrections, discuss the reasons behind them, and examine how these adjustments enhance clarity, thereby cultivating what we call “interactive language mastery.” For organization and structure, teachers might assign collaborative tasks in which students first use GenAI to generate an initial outline, then engage in group discussions or peer reviews to critically evaluate and refine these outlines to better meet writing goals. In addressing detail, specificity, and task relevance, teachers can assign tasks that require students to expand on GenAI-generated content by adding depth and context. Lessons on prompt engineering can further support this process by guiding students to craft precise prompts that steer GenAI outputs toward the desired tone and content. Such integrated instructional strategies can help students shift from conventional writing skills to the hybrid competencies required in the GenAI era, an approach that emphasizes using GenAI to enhance EFL students' writing adaptability rather than replacing their skills. This integration could both directly improve GenAI-assisted writing and indirectly improve it by fostering critical thinking, making students more capable and reflective writers.

Growth mindset did not show a significant direct effect, which may reflect a motivational dampening effect caused by the perceived ease and immediacy of GenAI-generated content. To counter this, teachers should adopt strategies that actively sustain motivation by re-emphasizing the value of human effort in conjunction with AI support. For example, teachers can create a classroom environment that visibly values human effort by explicitly recognizing and celebrating student-driven revisions of GenAI-generated drafts. This might include awarding points or praise for thoughtful edits, or showcasing before-and-after writing samples to highlight the impact of human input. In addition, teachers can introduce structured reflection activities such as reflective writing logs, where students record the challenges they faced, the ways they interacted with GenAI, and how their own decisions and effort contributed to improvement. Teachers may further guide this process by prompting students to analyze specific moments where their choices led to a better outcome than GenAI

alone, or where persistence and revision produced clearer or more accurate language. Another effective intervention is to design collaborative workshops in which students first use GenAI to generate initial drafts, and then work in pairs or small groups to critique, revise, and improve the output. In these sessions, teachers can circulate and offer targeted feedback, helping students see the value of critical thinking and sustained effort in shaping high-quality writing, even in the presence of AI tools. Over time, such practices help reinforce a growth-oriented classroom culture where GenAI is positioned as a resource for learning and creativity, not a shortcut or replacement for active skill development.

Moreover, the findings highlight important ethical concerns regarding GenAI use in writing instruction, particularly the risk of over-dependence and the erosion of students' personal voice and creativity. To address these concerns, instructional practices should establish clear boundaries between GenAI assistance and students' original contributions. Teachers can begin by developing explicit guidelines and rubrics that define permissible uses, for instance, allowing GenAI for idea expansion or language refinement while requiring students to formulate arguments and maintain authorial voice independently. To foster transparency and accountability, students may be asked to keep process logs that trace their writing trajectory, including initial drafts without GenAI input, subsequent revisions incorporating GenAI suggestions, and reflective annotations justifying these revisions. Peer review sessions or class discussions can further enhance this process by inviting students to compare GenAI-assisted and self-written drafts, evaluate their originality and coherence, and reflect on the value of human agency in writing. Finally, academic integrity instruction should be embedded throughout, addressing issues such as attribution, plagiarism, and responsible use. These integrated measures promote a pedagogical environment where GenAI serves as a scaffold reinforcing students' critical awareness, ethical responsibility, and ownership of the EFL writing process.

9 Conclusion

Building on an SCT-based framework, this study proposed and tested a model that conceptualizes EFL students as agentic learners who dynamically coordinate their linguistic, cognitive, motivational, and technological resources to regulate behavior and engage effectively in GenAI-assisted EFL writing. Specifically, five key constructs—self-regulation, critical thinking, growth mindset, GenAI literacy, and general writing competence—were examined for their influence on GenAI use. The research findings revealed: 1) Self-regulation, critical thinking, GenAI literacy, and general writing competence all significantly contributed to effective GenAI use in writing. Self-regulation and critical thinking emerged as the strongest predictors, followed by GenAI literacy, while general writing competence had a lower but still significant impact. 2) Growth mindset did not have a direct effect on GenAI use but indirectly influenced it through critical thinking and self-regulation. 3) General writing competence significantly predicted both growth mindsets and critical thinking, while critical thinking was a key determinant of self-regulation and GenAI literacy. Overall, these results suggest that cognitive factors are primary drivers of effective GenAI use

in writing, with GenAI literacy serving a supportive role. Unlike traditional perspectives that emphasized writing competence and motivation as central, these factors now have a less direct influence on GenAI-assisted environments. Together, these results highlight a shift toward a more integrated perspective that values cognitive engagement, technological literacy, and dynamic student-GenAI interaction. Theoretical and pedagogical implications are also discussed, underscoring the need to cultivate cognitive skills, embed GenAI literacy in instruction, reconceptualize writing competence, foster adaptive motivation, and address ethical considerations in GenAI-assisted EFL writing.

However, several limitations should be acknowledged. First, the study was conducted at a single university with Chinese EFL students, which limits the generalizability of the findings. Future research should include broader and more diverse samples, ideally through cross-cultural comparative studies, to enhance external validity. Second, the study relied predominantly on self-reported questionnaire data. Qualitative methods, such as interviews or classroom observations, could help validate the findings. Third, although the SCT-based model integrated key constructs, some hypotheses were not supported, suggesting that additional psychological and sociological variables may be necessary to fully explain GenAI-assisted writing dynamics. Future studies should consider incorporating factors such as technology self-efficacy, facilitating conditions, teacher support, or awareness of GenAI ethics in writing. These variables could enrich the discussion on how students develop a responsible and adaptive approach to GenAI-assisted writing. Finally, while this study employed PLS-SEM to model linear relationships and latent constructs, it did not capture possible combinatorial or configurational effects among variables. Given the increasing recognition of complexity in educational data, future studies could complement SEM with fuzzy-set Qualitative Comparative Analysis (fsQCA) to explore multiple causal pathways leading to effective GenAI use.

In conclusion, based on SCT, this study enhances our understanding of how EFL students, as agentic learners, coordinate cognitive, motivational, technological, and writing ability factors to engage effectively with GenAI in EFL writing. We hope these findings will inspire future research to develop more effective integration strategies for GenAI in writing instruction. Such efforts will contribute to the continued evolution of EFL writing education in the GenAI era.

Appendix 1: List of abbreviations

Abbreviation	Full Term
GenAI	Generative Artificial Intelligence
EFL	English as a Foreign Language
PLS-SEM	Partial Least Squares Structural Equation Modeling
SCT	Social Cognitive Theory

Authors' contributions Sha Tian designed the study and drafted and revised the manuscript. Han Xiang conducted the experiment and collected and analyzed the data.

Data availability The data used to support the findings are available from the corresponding author upon reasonable request. Some data are not available due to confidentiality restrictions.

Declarations

Competing interests The authors declare that they have no competing interests.

Informed consent Verbal consent was sought from the participants.

Declaration of Ggenerative AI and AI-assisted technologies in the writing process ChatGPT was only used in the writing process to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

References

- Alagozlu, N. (2007). Critical thinking and voice in EFL writing. *Asian EFL Journal*, 9(3), 118–136.
- Allen, T. J., & Mizumoto, A. (2024). ChatGPT over my friends: Japanese English-as-a-Foreign-Language learners' preferences for editing and proofreading strategies. *RELC Journal*, 00336882241262533. <https://doi.org/10.1177/00336882241262533>
- Bandura, A. (1986). *Social foundations of thought & action: A social cognitive theory*. Englewood Cliffs, NJ Prentice Hall.
- Bandura, A. (2001). Social cognitive theory: An agentic perspective. *Annual Review of Psychology*, 52, 1–26. <https://doi.org/10.1146/annurev.psych.52.1.1>
- Bandura, A. (2020). Social cognitive theory: An agentic perspective. *Psychology: The Journal of the Hellenic Psychological Society*, 12(3), 313–333. https://doi.org/10.12681/psy_hps.23964
- Baranowski, T., Perry, C., & Parcel, G. S. (2002). How individuals, environments, and health behavior interact. In K. Glanz, B. K. Rimer, & F. M. Lewis (Eds.), *Health Behavior and Health Education: Theory, Research, and Practice* (3rd ed., pp. 165–184). Jossey-Bass.
- Barrot, J. S. (2023). Using ChatGPT for second language writing: Pitfalls and potentials. *Assessing Writing*, 57, 100745. <https://doi.org/10.1016/j.asw.2023.100745>
- Boudouaia, A., Mouas, S., & Kouider, B. (2024). A study on ChatGPT-4 as an innovative approach to enhancing English as a foreign language writing learning. *Journal of Educational Computing Research*, 07356331241247465. <https://doi.org/10.1177/0735633124124746>
- Bucol, J. L., & Sangkawong, N. (2024). Exploring ChatGPT as a writing assessment tool. *Innovations in Education and Teaching International*, 1–16. <https://doi.org/10.1080/14703297.2024.2363901>
- Cadwell, P., Castilho, S., O'Brien, S., & Linda, M. (2016). Human factors in machine translation and post-editing among institutional translators. *Translation Spaces*, 5(2), 222–243. <https://doi.org/10.1075/ts.5.2.04cad>
- Cai, Q., Chen, W., Wang, M., & Di, K. (2025). The impact of self-determined efficacy on university student's environmental conservation intentions: An SEM-ANN exploration. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-024-05923-5>
- Carroll, D. W. (2007). Patterns of student writing in a critical thinking course: A quantitative analysis. *Assessing Writing*, 12(3), 213–227. <https://doi.org/10.1016/j.asw.2008.02.001>
- Cheong, C. M., Yao, Y., & Zhang, J. (2023). Growth mindset and emotions in tandem: Their effects on L2 writing performance based on writers' proficiency levels. *Assessing Writing*, 58, 100785. <https://doi.org/10.1016/j.asw.2023.100785>
- Chin, W. W. (1998). The partial least squares approach to structural equation modeling. In G. A. Marcoulides (Ed.), *Modern methods for business research* (pp. 295–336). Lawrence Erlbaum Associates.
- Chomsky, N., Roberts, I., & Watumull, J. (2023). Noam Chomsky: The false promise of ChatGPT. *The New York Times*, 8.
- Cole, K. L. (2024). Cultivating a growth mindset: A critical approach to teaching writing in the communication classroom. *Communication Teacher*. <https://doi.org/10.1080/17404622.2024.2399161>

- Condon, W., & Kelly-Riley, D. (2004). Assessing and teaching what we value: The relationship between college-level writing and critical thinking abilities. *Assessing Writing*, 9(1), 56–75. <https://doi.org/10.1016/j.asw.2004.01.003>
- Day, T. (2023). A preliminary investigation of fake peer-reviewed citations and references generated by ChatGPT. *The Professional Geographer*, 75(6), 1024–1027. <https://doi.org/10.1080/00330124.2023.2190373>
- Dewi, U. (2022). Grammarly as automated writing evaluation: Its effectiveness from EFL students' perceptions. *Lingua Cultura*, 16(2), 155–161. <https://doi.org/10.21512/lc.v16i2.8315>
- Di, K., Chen, W., Shi, Q., Cai, Q., & Liu, S. (2024). Analysing the impact of coupled domestic demand dynamics of green and low-carbon consumption in the market based on SEM-ANN. *Journal of Retailing and Consumer Services*, 79, 103856. <https://doi.org/10.1016/j.jretconser.2024.103856>
- Ding, L., & Zou, D. (2024). Automated writing evaluation systems: A systematic review of Grammarly, Pigai, and Criterion with a perspective on future directions in the age of generative artificial intelligence. *Education and Information Technologies*, 1–53. <https://doi.org/10.1007/s10639-023-12402-3>
- Dweck, C. S. (2006). *Mindset: The new psychology of success*. Random House.
- Dweck, C. S., & Leggett, E. L. (1988). A social-cognitive approach to motivation and personality. *Psychological Review*, 95(2), 256. <https://doi.org/10.1037/0033-295X.95.2.256>
- Ebadi, S., & Rahimi, M. (2019). Mediating EFL learners' academic writing skills in online dynamic assessment using Google Docs. *Computer Assisted Language Learning*, 32(5–6), 527–555. <https://doi.org/10.1080/09588221.2018.1527362>
- Elder, L., & Paul, R. (2004). Critical thinking and the art of close reading (Part II). *Journal of Developmental Education*, 27(3), 36–37.
- Eley, E. G. (1953). *An analysis of writing competence*. Dissertation. The University of Chicago.
- Facione, P. (1990). Critical thinking: A statement of expert consensus for purposes of educational assessment and instruction (The Delphi Report).
- Fatima, M., & van der Loo, J. (2024). *Assessing the impact of GenAI-assisted writing on grade attribution: The moderating role of self-efficacy in writing skill*. Master's Thesis. Tilburg University.
- FengTeng, M. (2024). A systematic review of ChatGPT for English as a Foreign Language writing: Opportunities, challenges, and recommendations. *International Journal of TESOL Studies*, 6(3), 36–57. <https://doi.org/10.58304/ijts.20240304>
- Fitria, T. N. (2021). QuillBot as an online tool: Students' alternative in paraphrasing and rewriting of English writing. *Englisia: Journal of Language, Education, and Humanities*, 9(1), 183–196. <https://doi.org/10.22373/ej.v9i1.10233>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.1177/002224378101800104>
- Gabrielatos, C. (2002). *EFL writing: Product and process*. Costas Gabrielatos.
- Garson, G. D. (2016). *Partial least squares: Regression & structural equation models*. Statistical Associates Blue Book Series.
- Gezmiş, N. (2020). Difficulties of students in process writing approach. *Journal of Language and Linguistic Studies*, 16(2), 565–579. <https://doi.org/10.17263/jlls.759249>
- Ghafouri, M., Hassaskhah, J., & Mahdavi-Zafarghandi, A. (2024). From virtual assistant to writing mentor: Exploring the impact of a ChatGPT-based writing instruction protocol on EFL teachers' self-efficacy and learners' writing skill. *Language Teaching Research*, 13621688241239764. <https://doi.org/10.1177/1362168824123976>
- Gilburt, I. (2023). A machine in the loop: The peculiar intervention of artificial intelligence in writer's block. *New Writing*, 21(1), 26–37. <https://doi.org/10.1080/14790726.2023.2223176>
- Gozali, I., Wijaya, A. R. T., Lie, A., Cahyono, B. Y., & Suryati, N. (2024). ChatGPT as an Automated Writing Evaluation (AWE) Tool: Feedback Literacy Development and AWE Tools' Integration Framework. *JALT CALL Journal*, 20(1), n1. <https://doi.org/10.29140/jaltcall.v20n1.1200>
- Guo, Y., & Lee, D. (2023). Leveraging ChatGPT for enhancing critical thinking skills. *Journal of Chemical Education*, 100(12), 4876–4883.
- Guo, K., & Wang, D. (2024). To resist it or to embrace it? Examining ChatGPT's potential to support teacher feedback in EFL writing. *Education and Information Technologies*, 29, 8435–8463. <https://doi.org/10.1007/s10639-023-12146-0>
- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2006). *Multivariate data analysis* (6th ed.). Prentice Hall.

- Hair, J. F., Ringle, C. M., & Sarstedt, M. (2013). Partial least squares structural equation modeling: Rigorous applications, better results, and higher acceptance. *Long Range Planning*, 46(1/2), 1–12. <https://doi.org/10.1016/j.lrp.2013.01.001>
- Halpern, D. F. (2013). *Thought and knowledge: An introduction to critical thinking*. Psychology Press.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43, 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Hoang, D. T. N., & Hoang, T. (2024). Enhancing EFL students' academic writing skills in online learning via Google Docs-based collaboration: A mixed-methods study. *Computer Assisted Language Learning*, 37(7), 1504–1526. <https://doi.org/10.1080/09588221.2022.2083176>
- Hsiao, C. H., & Tang, K. Y. (2024). Beyond acceptance: An empirical investigation of technological, ethical, social, and individual determinants of GenAI-supported learning in higher education. *Education and Information Technologies*. <https://doi.org/10.1007/s10639-024-13263-0>
- Hu, J., & Gao, X. (2018). Self-regulated strategic writing for academic studies in an English-medium-instruction context. *Language and Education*, 32(1), 1–20. <https://doi.org/10.1080/09500782.2017.1373804>
- Huang, J., & Mizumoto, A. (2024). Examining the effect of generative AI on students' motivation and writing self-efficacy. *Digital Applied Linguistics*, 1, 102324–102324. <https://doi.org/10.29140/dal.v1.102324>
- Hyland, K. (2003). Writing and teaching writing. *Second Language Writing*, 1(2), 1–30. <https://doi.org/10.1017/9781108693974.001>
- Jin, F., Lin, C. H., & Lai, C. (2025). Modeling AI-assisted writing: How self-regulated learning influences writing outcomes. *Computers in Human Behavior*, 165, 108538. <https://doi.org/10.1016/j.chb.2024.108538>
- Kaplan, A., Neuber, A., & Garner, J. K. (2019). An identity systems perspective on high ability in self-regulated learning. *High Ability Studies*, 30(1–2), 53–78. <https://doi.org/10.1080/13598139.2019.1568830>
- Kawinkoonlasate, P. (2019). *A comparative study of e-writing and traditional writing classroom to improve English writing ability and motivate autonomous learning of Thai EFL learners*. Huachiew Chalermprakiet University.
- Khalil, Z. M. (2018). EFL students' perceptions towards using Google Docs and Google Classroom as online collaborative tools in learning grammar. *Applied Linguistics Research Journal*, 2(2), 33–48. <https://doi.org/10.14744/alrj.2018.47955>
- Kim, J., Yu, S., Lee, S. S., & Detrick, R. (2025). Students' prompt patterns and its effects in AI-assisted academic writing: Focusing on students' level of AI literacy. *Journal of Research on Technology in Education*, 1–18. <https://doi.org/10.1080/15391523.2025.2456043>
- Kitjaroonchai, N., & Suppasetseree, S. (2021). Online collaborative writing via Google Docs: Case studies in the EFL classroom. *Journal of Language Teaching and Research*, 12(6), 922–934. <https://doi.org/10.17507/jltr.1206.08>
- Kline, R. B. (2005). *Principles and practice of structural equation modeling*. Guilford Publications.
- Lee, S. M. (2020). The impact of using machine translation on EFL students' writing. *Computer Assisted Language Learning*, 33(3), 157–175. <https://doi.org/10.1080/09588221.2018.1553186>
- Lee, J. M. (2023). A study on the data literacy education in the library of the Chat GPT, generative AI era. *Journal of the Korean Society for Library and Information Science*, 57(3), 303–323.
- Li, L., & Kim, M. (2024). It is like a friend to me: Critical usage of automated feedback systems by self-regulating English learners in higher education. *Australasian Journal of Educational Technology*, 40(1), 1–18. <https://doi.org/10.14742/ajet.8821>
- Liu, Z. M., Hwang, G. J., Chen, C. Q., Chen, X. D., & Ye, X. D. (2024). Integrating large language models into EFL writing instruction: effects on performance, self-regulated learning strategies, and motivation. *Computer Assisted Language Learning*, 1–25. <https://doi.org/10.1080/09588221.2024.2389923>
- Marzuki Widiati, U., Rusdin, D., Darwin, & Indrawati, I. (2023). The impact of AI writing tools on the content and organization of students' writing: EFL teachers' perspective. *Cogent Education*, 10(2), 1–17. <https://doi.org/10.1080/2331186X.2023.2236469>
- Mehta, S. R., & Al-Mahrooqi, R. (2015). Can thinking be taught? Linking critical thinking and writing in an EFL context. *RELJ Journal*, 46(1), 23–36.
- Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2), 100050. <https://doi.org/10.1016/j.rm.al.2023.100050>

- Nazarov, A., & Kuvandikova, D. (2023). Developing writing competence. *Current Approaches and New Research in Modern Sciences*, 2(5), 71–75.
- Nickerson, C. (2024). Albert Bandura's social cognitive theory. *Simply Psychology*, 2.
- Onoda, S. (2012). *Self-regulation and its relation to motivation and proficiency*. Temple University.
- Punar Özgelik, N., & Yangın Ekşi, G. (2024). Cultivating writing skills: The role of ChatGPT as a learning assistant—a case study. *Smart Learning Environments*, 11(1), 10. <https://doi.org/10.1186/s40561-024-00296-8>
- Putnam, D. (2001). Authentic writing using online resources: Selling our words in the community. *The English Journal*, 90(5), 102–106. <https://doi.org/10.2307/821862>
- Ridley, M., & Pawlick-Potts, D. (2021). Algorithmic literacy and the role for libraries. *Information Technology and Libraries*, 40(2). <https://doi.org/10.6017/ital.v40i2.12963>
- Sánchez-Peña, D., & Chapetón, C. M. (2018). Fostering the development of written argumentative competence in ELT from a critical literacy approach. *Revista Colombiana de Educación*, 75, 159–183. <https://doi.org/10.17227/rce.num75-8107>
- Santangelo, T., Harris, K., & Graham, S. (2016). Self-regulation and writing. *Handbook of Writing Research*, 2, 174–193.
- Schunk, D. H., & DiBenedetto, M. K. (2020). Motivation and social cognitive theory. *Contemporary Educational Psychology*, 60, Article 101832. <https://doi.org/10.1016/j.cedpsych.2019.101832>
- Shen, X., & Teng, M. F. (2024). Three-wave cross-lagged model on the correlations between critical thinking skills, self-directed learning competency and AI-assisted writing. *Thinking Skills and Creativity*, 52, 101524. <https://doi.org/10.1016/j.tsc.2024.101524>
- Shiri, A. (2023). ChatGPT and academic integrity. *Information Matters*, 3(2).
- Snyder, L. G., & Snyder, M. J. (2008). Teaching critical thinking and problem solving skills. *The Journal of Research in Business Education*, 50(2), 90.
- Song, C., & Song, Y. (2023). Enhancing academic writing skills and motivation: Assessing the efficacy of ChatGPT in AI-assisted language learning for EFL students. *Frontiers in Psychology*, 14, 1260843. <https://doi.org/10.3389/fpsyg.2023.1260843>
- Sullivan, M., McAuley, M., Degiorgio, D., & McLaughlan, P. (2024). Improving students' generative AI literacy: A single workshop can improve confidence and understanding. *Journal of Applied Learning and Teaching*, 7(2). <https://doi.org/10.37074/jalt.2024.7.2.7>
- Taskiran, A., Yazici, M., & Erdem Aydin, I. (2024). Contribution of automated feedback to the English writing competence of distance foreign language learners. *E-Learning and Digital Media*, 21(1), 24–41. <https://doi.org/10.1177/20427530221139579>
- Tu, Y. F. (2024). Roles and functionalities of ChatGPT for students with different growth mindsets. *Educational Technology & Society*, 27(1), 198–214. [https://doi.org/10.30191/ETS.202401_27\(1\).TP01](https://doi.org/10.30191/ETS.202401_27(1).TP01)
- Umamah, A., & Cahyono, B. Y. (2022). EFL university students' use of online resources to facilitate self-regulation for writing. *Computer-Assisted Language Learning Electronic Journal*, 23(1), 108–124.
- Wang, H., & Dang, A. (2024). Enhancing L2 writing with generative ai: A systematic review of pedagogical integration and outcomes. Preprint: [https://doi.org/10.13140/RG.2\(19572.16005\)](https://doi.org/10.13140/RG.2(19572.16005))
- Weigle, S. C. (2007). Teaching writing teachers about assessment. *Journal of Second Language Writing*, 16(3), 194–209. <https://doi.org/10.1016/j.jslw.2007.07.004>
- Weissman, J. (2023). ChatGPT is a plague upon education (opinion). Inside Higher Ed. <https://www.insidehighered.com/views/2023/02/09/chatgpt-plague-uponeducation-opinion>
- Werdiningsih, I., Marzuki, Indrawati, I., Rusdin, D., Ivone, F. M., Basthomi, Y., & Zulfahreza. (2024). Revolutionizing EFL writing: Unveiling the strategic use of ChatGPT by Indonesian master's students. *Cogent Education*, 11(1), 2399431. <https://doi.org/10.1080/2331186X.2024.2399431>
- Wetzels, M., Odekerken-Schröder, G., & Van Oppen, C. (2009). Using PLS path modeling for assessing hierarchical construct models: Guidelines and empirical illustration. *MIS Quarterly*, 177–195. <https://doi.org/10.2307/20650284>
- Xu, J., & Wang, Y. (2022). The differential mediating roles of ideal and ought-to L2 writing selves between growth mindsets and self-regulated writing strategies. *System*, 110, 102900. <https://doi.org/10.1016/j.system.2022.102900>
- Yang, Y., & Wang, X. (2019). Modeling the intention to use machine translation for student translators: An extension of technology acceptance model. *Computers & Education*, 133, 116–126. <https://doi.org/10.1016/j.compedu.2019.01.015>
- Yao, Y., Yu, S., Zhu, X., Zhu, S., & Pang, W. (2024). Examining the mindsets toward students' writing competence and feedback-giving practices of Chinese university English writing teachers. *Studies in Educational Evaluation*, 80, 101318. <https://doi.org/10.1016/j.stueduc.2023.101318>

- Yasuda, S. (2011). Genre-based tasks in foreign language writing: Developing writers' genre awareness, linguistic knowledge, and writing competence. *Journal of Second Language Writing*, 20(2), 111–133. <https://doi.org/10.1016/j.jslw.2011.03.001>
- Yu, H. (2023). Reflection on whether chat GPT should be banned by academia from the perspective of education and teaching. *Frontiers in Psychology*, 14, 1181712. <https://doi.org/10.3389/fpsyg.2023.1181712>
- Zhai, X. (2023). Chatgpt and AI: The game changer for education. In Zhai, X.(2023). *ChatGPT: Reforming Education on Five Aspects*. Shanghai Education, 16–17.
- Zhao, X., Cox, A., & Cai, L. (2024). ChatGPT and the digitisation of writing. *Humanities and Social Sciences Communications*, 11, 482. <https://doi.org/10.1057/s41599-024-02904-x>
- Zhou, X., Teng, D., & Al-Samarraie, H. (2024). The mediating role of generative AI self-regulation on students' critical thinking and problem-solving. *Education Sciences*, 14(12), 1302. <https://doi.org/10.3390/educsci14121302>
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory into Practice*, 41(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2
- Zimmerman, B. J., & Kitsantas, A. (2014). Comparing students' self-discipline and self-regulation measures and their prediction of academic achievement. *Contemporary Educational Psychology*, 39(2), 145–155. <https://doi.org/10.1016/j.cedpsych.2014.03.004>

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